

TriboAI Launches Revolutionary Smart Sensing System to Transform Industrial Maintenance

Southampton, UK, January 13, 2026.

TriboAI, a new spinout from the University of Southampton, today announced the commercial release of its flagship Machinery Condition Monitoring System. This groundbreaking technology combines advanced electrostatic sensing with proprietary machine learning algorithms to detect early signs of bearing wear and lubricant breakdown, offering a transformative solution for industries ranging from wind energy to rail transport.

The release marks a significant milestone for the company, which has transitioned from academic research at the National Centre for Advanced Tribology at Southampton (nCATS) to a commercial entity backed by decades of expertise. The system addresses two critical industrial challenges: The premature replacement of bearings, often discarded while still functional to avoid catastrophic failure, and the reluctance to adopt sustainable plant-based lubricants due to performance uncertainty. In doing so, the system also supports the UK's Net Zero 2050 and Digital Industry strategies.

A "Smart Skin" for Machinery

TriboAI's first product serves as a "smart skin" for critical machinery. Unlike traditional vibration monitoring, which often detects faults only after damage has occurred, TriboAI's electrostatic sensors detect the earliest disturbances in the lubricant film and surface interactions.

"Preventative maintenance has historically been conservative and expensive, forcing operators to replace components far too early just to be safe," said Professor Robert Wood, Founding Director of TriboAI and a veteran of 30 years in electrostatic sensor research. "Our technology changes this paradigm. By 'listening' to the tribological conditions inside the bearing in real-time, we allow operators to safely extend asset life and maximize utility without the risk of unexpected failure." The team hope that fatal accidents, such as the 2018 Leicester helicopter crash that killed 5, can be avoided in future by adopting the technology.

The system integrates seamlessly into existing infrastructure, using non-intrusive sensors that feed data into an AI-powered analytics platform. This platform uses outlier detection methods to identify abnormal events that signify potential failures, pinpointing specific failure modes with high precision.

Enabling the Green Transition

The launch comes at a pivotal moment as industries strive to meet the UN and UK's sustainability goals, including SDG 6 (clean water) and SDG 7 (clean energy). While oil-based lubricants can be tremendously damaging to the environment, operators have hesitated to switch to plant-based "green" lubricants for critical safety applications due to a lack of long-term performance and stability data.

TriboAI removes this barrier. "Our sensors provide the real-time assurance operators need to switch to sustainable lubricants," said Professor Nick Harris, co-founder and expert in sensor technology. "If we can give industry the confidence to move away from fossil-fuel-based oils, the environmental impact will be immense, preventing groundwater contamination and reducing industrial carbon footprints."

Validated Impact and Market Potential

The impact of this technology is far-reaching. According to external market analysis, the predictive maintenance market is projected to reach \$60 billion by 2030. TriboAI's solution is particularly vital for machinery which will enable the UK to meet its 2050 Net Zero decarbonisation goals, like offshore wind turbines, where replacing a bearing is dangerous, costly, and weather-dependent.

Key Benefits of the TriboAI system

Reduced Operational Costs: Predictive maintenance can deliver cost reductions of 18-25% compared to standard approaches.

Environmental Sustainability: Extending bearing life reduces manufacturing waste, while enabling green lubricants protects ecosystems.

Enhanced Safety: The system protects both infrastructure and human life, for example by catching potential train derailment risks or failures in power generation early.

Commercial Availability

The TriboAI Machinery Condition Monitoring System is available immediately for pilot deployment in key sectors including renewable energy, rail, and manufacturing. The company is currently engaged with major industrial partners to integrate the technology into next-generation assets, and has a signed commitment from SKF, the largest European bearing manufacturer.

"We are not just selling a sensor; we are providing peace of mind and a pathway to sustainability," added the commercial leadership team. "Whether it is keeping a wind turbine spinning or ensuring a train reaches its destination safely, TriboAI is making future machines smart."

About TriboAI

TriboAI is a University of Southampton spinout company dedicated to making future machines smart through advanced tribology and sensor technology. Born from the world-renowned National Centre for Advanced Tribology at Southampton (nCATS), TriboAI combines over 20 years of research with cutting-edge AI to provide predictive maintenance solutions that optimize efficiency, reduce downtime, and extend equipment life.

For more information, please visit <https://triboai.co.uk>

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